

Adelowo Adebangi Oluwatimileyin

Data Scientist · Machine Learning Engineer · Applied Mathematician
adelowooluwatimileyin@gmail.com · +39 388 751 5527 · L'Aquila, Italy
github.com/AdebangiAdelowo · adebanjioluwatimileyin.github.io

PROFILE

Applied Mathematician and Data Scientist with 3+ years of professional experience building predictive models, operational analytics, signal-processing pipelines, and production ML services. Combines strong foundations in mathematical modeling, optimization, numerical methods, and statistics with hands-on experience in Python, SQL, PyTorch, scikit-learn, FastAPI, and Docker. Additional project work spans LLM systems, scientific machine learning, computer vision, and computational modeling.

PROFESSIONAL EXPERIENCE

- | | |
|---------------------|--|
| Dec 2024 – Present | Data Scientist <i>Intuos Srl</i> , L'Aquila, Italy <ul style="list-style-type: none">Develop aircraft flight-phase classifiers from operational telemetry; F1 score of 0.98 on an evaluation set.Develop FFT-based audio signal-processing pipelines to analyze aircraft-engine recordings in the frequency domain, identify dominant frequency components, estimate engine RPM, and infer engine operating state across 1,500 telemetry/audio samples; validated RPM estimates against ground-truth measurements, achieving $R^2 = 0.80$.Build automated flight-telemetry workflows with a reported 30% processing-speed improvement and 80% less manual data preparation.Deliver REST APIs and a full-stack dashboard exposing model predictions and engineering metrics; over 200 users. |
| Apr 2023 – Nov 2024 | Machine Learning Engineer <i>N-and Italia Srl</i> , Italy <ul style="list-style-type: none">Developed delivery-failure prediction models using over 5 million transaction records from approximately 150 vending machines, with reported F1 of 0.85.Analyzed transactional and telemetry patterns to investigate vending-machine reliability and operational failure drivers.Built ETL and KPI-monitoring workflows with reported 85% faster reporting and monitoring coverage of over 1 million transaction records. |
| Jan 2023 – Mar 2023 | Operational Data Analyst <i>eResult Srl</i> , Cesena, Italy <ul style="list-style-type: none">Optimized SQL queries, reporting 45% faster execution, and automated recurring business-intelligence workflows.Transformed operational data into reproducible KPI reports, with a reported 75% reduction in reporting effort.Built automated, customized customer relationship management (CRM) platforms using SQL and Omniplace, its integrated software development platform, tailoring workflows and features for operational performance, efficiency, and versatility. |
| Apr 2022 – Oct 2022 | Digital Engineering Intern <i>Bridgestone EMIA</i> , Rome, Italy <ul style="list-style-type: none">Developed computational-geometry algorithms and improved data structures for tire cross-section simulation and finite-element meshing, with a reported 20% computational speed improvement.Implemented a closest-point utility for cubic Bézier curves to support geometry projection and meshing operations. |
| May 2022 – Aug 2022 | Machine Learning Intern <i>Fellowship.AI</i> , Remote <ul style="list-style-type: none">Integrated an InsetGAN-based whole-body image-generation pipeline (StyleGAN-Human with InsetGAN joint optimization) and built FastAPI inference serving for a Swift iOS application; reported evaluation on 500 generated samples and 2.3-second inference latency. |

TECHNICAL SKILLS

Programming and data: Python, SQL, C++, MATLAB, JavaScript, NumPy, SciPy, Pandas, PostgreSQL, IBM DB2

Machine learning: PyTorch, TensorFlow, scikit-learn, LightGBM, supervised learning, deep learning, feature engineering, model evaluation, time-series and signal processing

LLM and generative AI: LLMs, RAG, LangChain, OpenAI APIs, ChromaDB, NL-to-SQL, prompt engineering, embeddings, vector search

Engineering and MLOps: FastAPI, Docker, MLflow, Apache Spark, Databricks, Git/GitHub, ETL/data pipelines, REST APIs, AWS, Kubernetes, Terraform

Scientific computing: numerical methods, optimization, finite-element and pseudo-spectral methods, computational modeling, PDEs

SELECTED PROJECTS

INTUOS Flight Data Monitoring Dashboard: Full-stack platform using FastAPI, IBM DB2, React/Vite, and Docker for query-on-demand analysis of recorded flight telemetry. Serves ML-based flight-phase classifiers; deterministic logic drives safety alarms. Developed for internal flight-performance analysis at Intuos Srl.

LLM Engineering Portfolio: Implemented Medical RAG, Enterprise NL-to-SQL, and FinBERT pruning/distillation pipelines within a nine-project portfolio. Compressed FinBERT by 20% (110M to 88M parameters); distillation recovered accuracy lost through layer dropping. Evaluation supports relative compression effects, with likely overlap between the test split and baseline training data.

Aircraft Engine Monitoring via Audio DSP: Built FFT-based RPM and status estimation on a Raspberry Pi, transmitting results over UART to an ESP32. Tested on real Tecnam P92 / Rotax 912 engine recordings and analyzed harmonic-order ambiguity; validated RPM estimation against ground-truth measurements ($R^2 = 0.80$).

Optimal Mixing of Passive Scalars: Derived and implemented the Lin–Thiffeault–Doering scheme in a pseudo-spectral Python solver, reproducing the original thesis tables. Extended the resolution study to $N = 512$ with 48 verification tests; identified changing fit windows as the main cause of apparent exponent dependence on resolution, with approximately second-order convergence on fixed windows.

Burgers' Equation: ROM/FNO Comparison: Compared POD–Galerkin reduced-order models, DEIM hyper-reduction, an MLP, and a Fourier Neural Operator against a pseudo-spectral solver. Across 40 in-distribution cases, POD-ROM achieved 0.152% relative L^2 error versus 0.90% for FNO (five seeds). Testing an unseen initial-condition family exposed limited FNO transfer: 22.6% error versus 1.69% for POD-ROM.

Photoacoustic Reconstruction: Model-Based vs. Learned Inversion: Built a synthetic sparse-view imaging pipeline comparing time reversal, Tikhonov inversion, and learned reconstruction in PyTorch. Across 400 held-out images and five U-Net training seeds, refinement improved PSNR over calibrated time reversal by 4.9 dB at 16 sensors and 3.2 dB at 64 sensors. Tested sound-speed mismatch and distribution shift, identifying limitations on unseen thin vessel-like structures.

EDUCATION

2018 – 2021	M.Sc. Mathematical Engineering University of L'Aquila, Italy Thesis: <i>Bounds on Mixing of Passive Scalars Advected by Incompressible Energy- and Enstrophy-Constrained Flows and Anomalous Dissipation.</i> Relevant study: machine learning, scientific computing, optimization, numerical analysis, finite-element methods, and PDEs.
2013 – 2018	B.Sc. Mathematics Obafemi Awolowo University, Nigeria Thesis: <i>Application of Galerkin Finite Element Method to the One-Dimensional Stefan Problem.</i>

LEADERSHIP

Leadership: Founder and President, Inter-University Mathematics Club (2016–2018); Academic Director and peer tutor, NAMSS, Obafemi Awolowo University (2015–2018).